

# Rahul Dange

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## TECHNICAL SKILLS

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**Systems & Low-Level:** C, C++, x86-64 Assembly, Linux Kernel, POSIX Threads, Concurrency, IPC, Memory Management, GDB, Valgrind

**Languages & Paradigms:** Python, Java, TypeScript, JavaScript, SQL (PostgreSQL), Microservices Architecture, System Design, SOLID

**Frameworks & APIs:** React, Next.js, Node.js, Express, FastAPI, RESTful Architecture

**Infrastructure & Telemetry:** Linux, Bash/Shell, Git, Docker, Google Cloud (GCP), AWS, CI/CD (GitHub Actions), OpenTelemetry

## EXPERIENCE

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### Academic Tutor – COMP2017 : System Programming (Part-Time)

Sydney, Australia

*The University of Sydney*

*Feb 2026 – Present*

- Mentoring 200+ students in low-level system programming (C/C++), debugging volatile memory and concurrency faults via GDB/Valgrind.
- Managing EdStem query workflows and administrative tracking, strictly categorized under the GEN-EA CLAUSE 68 compliance framework.

### Full Stack Engineer (Part-Time)

Pune, India

*ChessVerst*

*Feb 2021 – Jul 2023*

- Architected scalable software solutions and full-stack features utilizing JavaScript (ES6+), TypeScript, and modern architectural patterns.
- Engineered decoupled backend microservices to optimize data transport protocols and ensure high-availability cross-platform performance.

### Associate Software Engineer (Part-Time)

Remote

*Calc Technologies*

*Mar 2022 – May 2022*

- Developed dynamic, responsive user interfaces leveraging React.js, enhancing platform usability and client-side rendering speeds.

## TECHNICAL PROJECTS

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### Geometric Storage Database | *ISO C11, Direct OS I/O, Advanced Math*

- Engineered an experimental  $\mathcal{O}(1)$  constant-time document store, bypassing traditional B-Trees by mapping 4D mathematical coordinates directly to physical disk offsets.

### Sentinel Daemon and Client CLI | *C, Assembly, macOS/Linux Internals*

- Developed a zero-dependency autonomous daemon executing continuous malware sweeps, memory intrusion auto-healing, and proactive RAM purging.
- Engineered dynamic background CPU throttling mechanisms to optimally prioritize and scale interactive hardware workloads.

### SpatialPress – Binary Compression Codec | *Algorithms, 64-bit Range Coding*

- Architected a high-performance, modular binary compression codec utilizing multi-dimensional coordinate interleaving via 4D Morton Z-order curves.
- Integrated 64-bit adaptive range coding algorithms to maximize spatial data compression density and decoding throughput.

### Spotlight – Social Interaction Platform | *React, Node.js, Express, MongoDB, Lerna*

- Built a scalable, full-stack interaction platform structured entirely within a centralized monorepo architecture for streamlined deployment pipelines.
- Engineered dynamic client interfaces with JavaScript/React while orchestrating asynchronous backend data transport using Node.js and MongoDB.

EDUCATION

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|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| <b>The University of Sydney</b><br><i>Master of Computer Science (Specialization in Software Engineering)</i>                                        | Sydney, Australia<br>Feb 2024 – Dec 2025 |
| <b>Nutan College of Engineering and Research (NCER)</b><br><i>Bachelor of Technology in Computer Science and Engineering (Affiliated with DBATU)</i> | Pune, India<br>Aug 2019 – Aug 2023       |

PUBLICATIONS

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| <b>Revolutionizing Student Engagement: Building a Cutting-Edge Interaction System</b><br><i>SSRN (Elsevier)   5th Intl. Conf. on Communication &amp; Information Processing (ICCIP-2023)</i>                                                                                                                                                                                                                                                            | Nov 2023 |
| <ul style="list-style-type: none"><li>• Authored a technical blueprint for a high-performance interaction platform utilizing the MERN stack and Next.js, analyzing Server-Side Rendering (SSR) to mitigate traditional MVC performance bottlenecks.</li><li>• Evaluated system response times under heavy concurrent loads, benchmarking optimized asset delivery pipelines and state management using TypeScript, Cloudinary, and Turbopack.</li></ul> |          |
| <b>Student Interaction System Using Modern Web Technologies</b><br><i>Scandinavian Journal of Information Systems, 35(1), 142–151</i>                                                                                                                                                                                                                                                                                                                   | Feb 2023 |
| <ul style="list-style-type: none"><li>• Co-authored research analyzing the integration and architectural implementation of modern web frameworks to optimize educational interaction platforms.</li><li>• Published findings on system scalability, web technology utilization, and platform performance in collaboration with academic researchers.</li></ul>                                                                                          |          |

VOLUNTEERING & LEADERSHIP

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| <b>Senior Student Advisor</b><br><i>Nutan College of Engineering and Research (NCER)</i>                                                                                                                                                                                                                                                                                                          | Pune, India<br>Jul 2022 – Jul 2023 |
| <ul style="list-style-type: none"><li>• Appointed by the Head of the CS Department (Dr. Sanjeevkumar Angadi) as a strategic mentor to facilitate the professional transition of junior cohorts into software engineering careers.</li><li>• Served as the primary administrative liaison between the student body and faculty leadership to proactively resolve academic roadblocks.</li></ul>    |                                    |
| <b>Primary Student Coordinator, NCER Hackathon 2022</b><br><i>Nutan College of Engineering and Research (NCER)</i>                                                                                                                                                                                                                                                                                | Pune, India<br>2022                |
| <ul style="list-style-type: none"><li>• Spearheaded the institution’s inaugural large-scale hackathon as the sole coordinator, representing and managing students across all four years of the CSE department.</li><li>• Provided rigorous technical mentorship to 8 cross-functional teams, guiding them through product architecture, technical roadblocks, and execution strategies.</li></ul> |                                    |
| <b>NCER++ LMS Maintainer</b><br><i>Nutan College of Engineering and Research (NCER)</i>                                                                                                                                                                                                                                                                                                           | Pune, India<br>Feb 2020 – Sep 2023 |
| <ul style="list-style-type: none"><li>• Engineered and scaled a foundational Java application into a comprehensive Learning Management System, delivering academic resources and unit-specific technical roadmaps.</li><li>• Transformed the platform into a centralized community hub to equip students with career-building information and industry-ready skills.</li></ul>                    |                                    |
| <b>Founder &amp; Coordinator, NCER Cyber Security Club</b><br><i>Nutan College of Engineering and Research (NCER)</i>                                                                                                                                                                                                                                                                             | Pune, India<br>Sep 2019 – Sep 2022 |
| <ul style="list-style-type: none"><li>• Established the institution’s first cybersecurity community focused on ethical hacking, digital safety, and hands-on application of real-world security concepts.</li><li>• Launched and managed the official NCER Cybersecurity Blog to disseminate technical write-ups and security research to the broader student body.</li></ul>                     |                                    |
| <b>Batch Representative, English Proficiency Initiative</b><br><i>Nutan College of Engineering and Research (NCER)</i>                                                                                                                                                                                                                                                                            | Pune, India<br>Aug 2019 – May 2020 |
| <ul style="list-style-type: none"><li>• Collaborated directly with institutional faculty to optimize teaching methodologies, successfully tailoring the curriculum for students from regional backgrounds.</li><li>• Mentored peers to bridge communication gaps, ensuring language proficiency was utilized effectively for professional career development and corporate placements.</li></ul>  |                                    |